

Review of SVP calibration for the well-log data

svpChange2 simulation audit

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1 Recommendation

For this data, the preferred calibrated configuration is

```
SVP(y, gamma = 84.85,  
    test = "MedianMoodCost", subtests = "right", cost = "gaussian")
```

The threshold is the empirical 0.99 quantile of the maximum Median–Mood statistic over every growing prefix under a dependence-preserving residual block bootstrap. This is exactly the one-segment criterion for **right**; it is not selected to maximize agreement with RFPOP. The recommendation changes the current application from **both** to **right**, removes the oracle segment count, and accounts for serial dependence.

2 Existing analysis and literature benchmark

The current application contains two incompatible oracle segment counts: 30 for Wilcoxon and 10 for Median–Mood. Its Mood threshold also uses a χ_K^2 quantile, although the Pearson statistic has one degree of freedom. Neither construction accounts for the substantial serial dependence remaining in the well-log data.

The external benchmark is the existing RFPOP fit with biweight loss, `lambda = 70`, and `lthreshold = 2`. This reproduces the application of Fearnhead and Rigall (2019), who use a threshold of twice the estimated noise scale and a penalty of 70 estimated noise variances. It gives the following internal changepoints:

[1] 1070 1526 1685 1866 2047 2409 2469 2531 2591 2768

RFPOP is treated as an external benchmark, not ground truth. In particular, using a first-difference MAD rather than the package application’s global MAD adds a short segment near positions 1034–1069. The 19-change analysis in Wilms, Killick, and Matteson (2022) is not a compatible gold standard: it analyzes a 1,000-observation subset with Gaussian PELT and emphasizes the instability of that segmentation.

Table 1: Sensitivity of the RFPOP benchmark to noise-scale estimation.

scale_method	scale	n_changepoints
package_global_mad	6150.12	10
first_difference_mad	2162.13	11

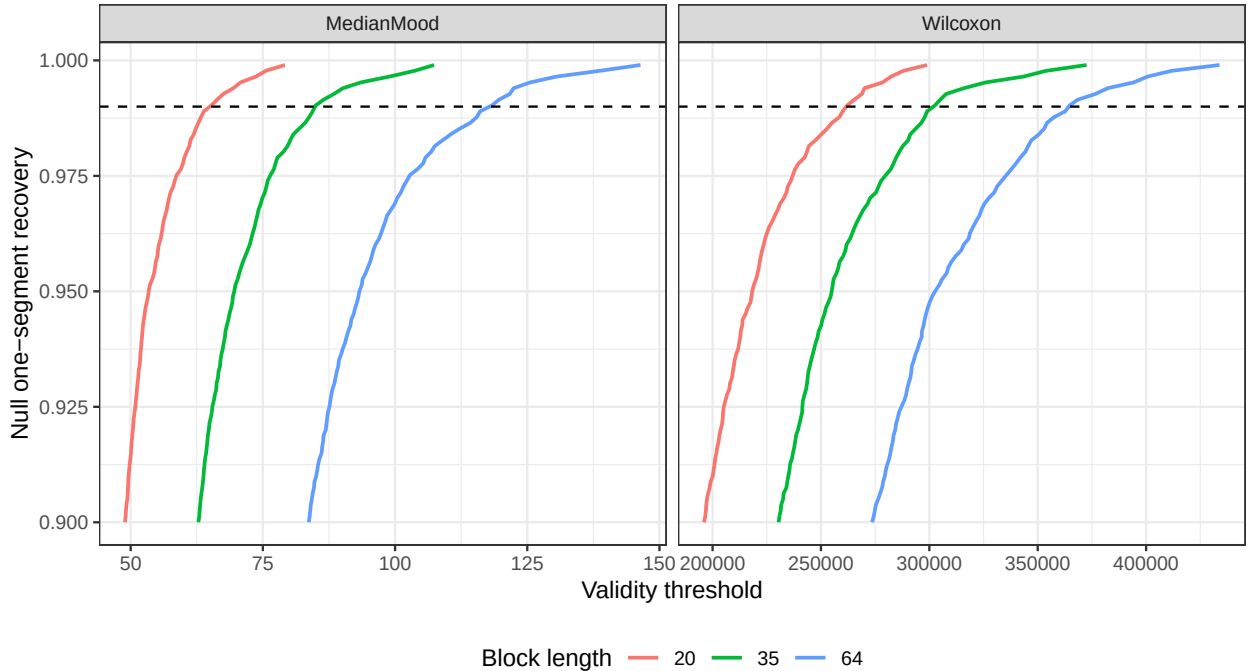
The global-MAD fit gives changes at 1070, 1526, 1685, 1866, 2047, 2409, 2469, 2531, 2591, 2768. The first-difference-MAD fit gives changes at 1034, 1069, 1526, 1683, 1866, 2046, 2408, 2468, 2531, 2591, 2768.

3 Null calibration

We subtract the median in each RFPOP segment and resample residual blocks that do not cross a reference boundary. The block length is the first lag followed by five residual autocorrelations inside $\pm 1.96/\sqrt{n}$; this gives 35. Calibration and independent validation use 5,000 series each, with block lengths 20 and 64 included as sensitivity checks. A null series is counted as correct only when the complete **right** run returns one segment.

Table 2: Block-bootstrap null calibration and independent validation.

test	block_length	gamma	calibration_recovery	validation_recovery	validation_lower	validation_upper
MedianMood	20	64.301	0.99	0.9896	0.9864	0.9921
Wilcoxon	20	261882.000	0.99	0.9908	0.9878	0.9931
MedianMood	35	84.850	0.99	0.9882	0.9848	0.9908
Wilcoxon	35	302231.000	0.99	0.9888	0.9855	0.9914
MedianMood	64	116.786	0.99	0.9898	0.9866	0.9922
Wilcoxon	64	363742.000	0.99	0.9896	0.9864	0.9921



The Wilcoxon scan is strongly inflated by the residual dependence. Its calibrated threshold is therefore large and retains only the most prominent transitions. The oracle- K Wilcoxon scaling used in the power experiment is not suitable for this real-data application.

The same threshold cannot be described as a **both** calibration. In that mode, an invalid interior candidate can remove the one-segment candidate even when all full-prefix tests pass. The direct stress check below therefore reports the actual **both** one-segment rate at the **right** threshold.

Table 3: Direct null stress check for both over 50 dependent bootstrap series.

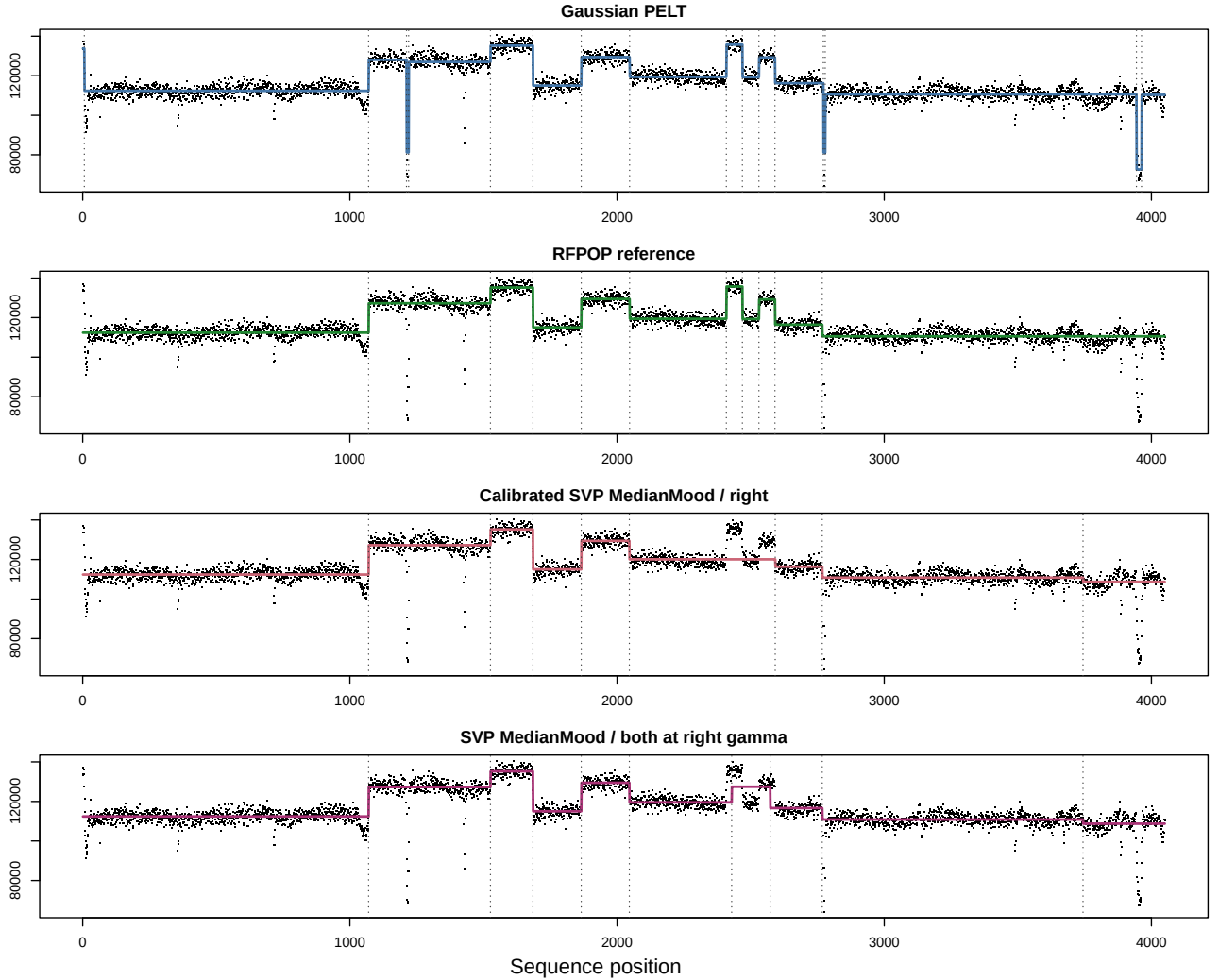
subtests	gamma	one_segment_recovery	lower	upper	reps
both	84.85	0.94	0.838	0.979	50

4 Observed segmentations

All comparisons below remove the terminal endpoint and use one-to-one matching within 10 observations. These scores measure agreement with RFPOP rather than accuracy against known geological boundaries.

Table 4: Observed-data agreement with the external RFPOP benchmark.

method	n_changepoints	matched	precision	recall	F1
Gaussian PELT	16	10	0.625	1.0	0.769
RFPOP reference	10	10	1.000	1.0	1.000
Legacy SVP Wilcoxon / both	33	6	0.182	0.6	0.279
Legacy SVP MedianMood / both	17	9	0.529	0.9	0.667
SVP Wilcoxon / right	2	2	1.000	0.2	0.333
SVP MedianMood / right	8	7	0.875	0.7	0.778
SVP MedianMood / both at right gamma	9	6	0.667	0.6	0.632



The calibrated `right` fit matches seven of the ten RFPOP boundaries and adds one SVP-only candidate near 3744. It merges several short neighboring segments in the 2400–2600 cluster; this is the cost of controlling false segmentation under the strongly dependent null. At the same threshold, `both` does not improve agreement because some clustered boundaries shift outside the matching tolerance. The legacy Mood and Wilcoxon settings over-segment substantially.

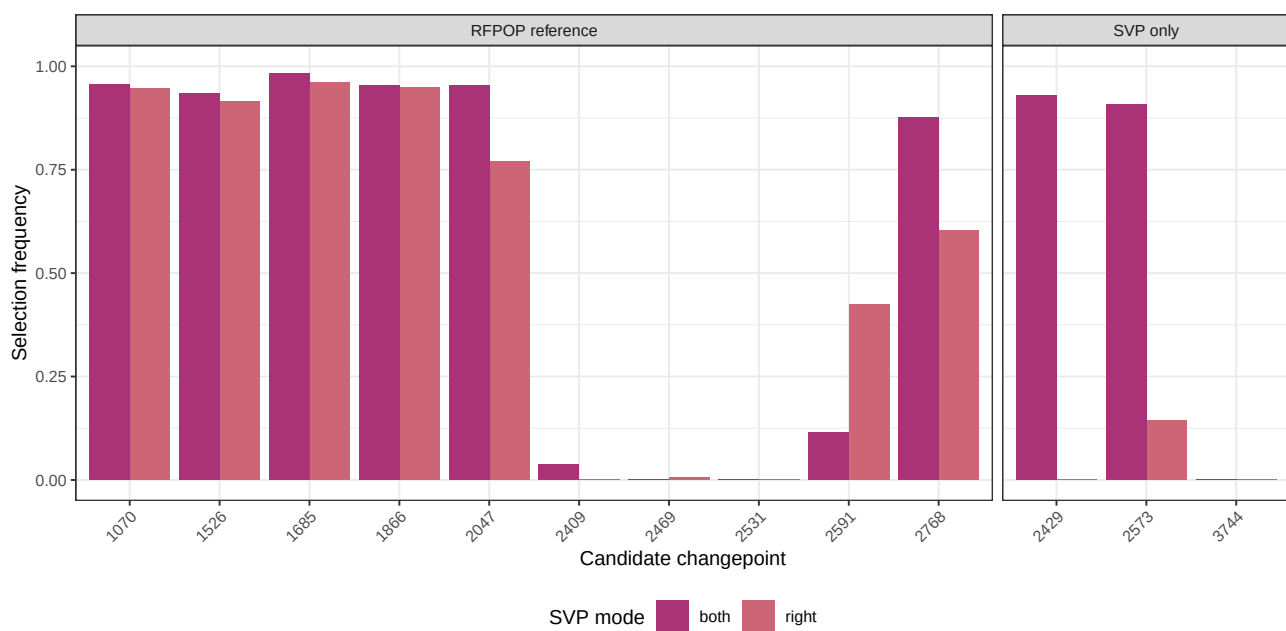
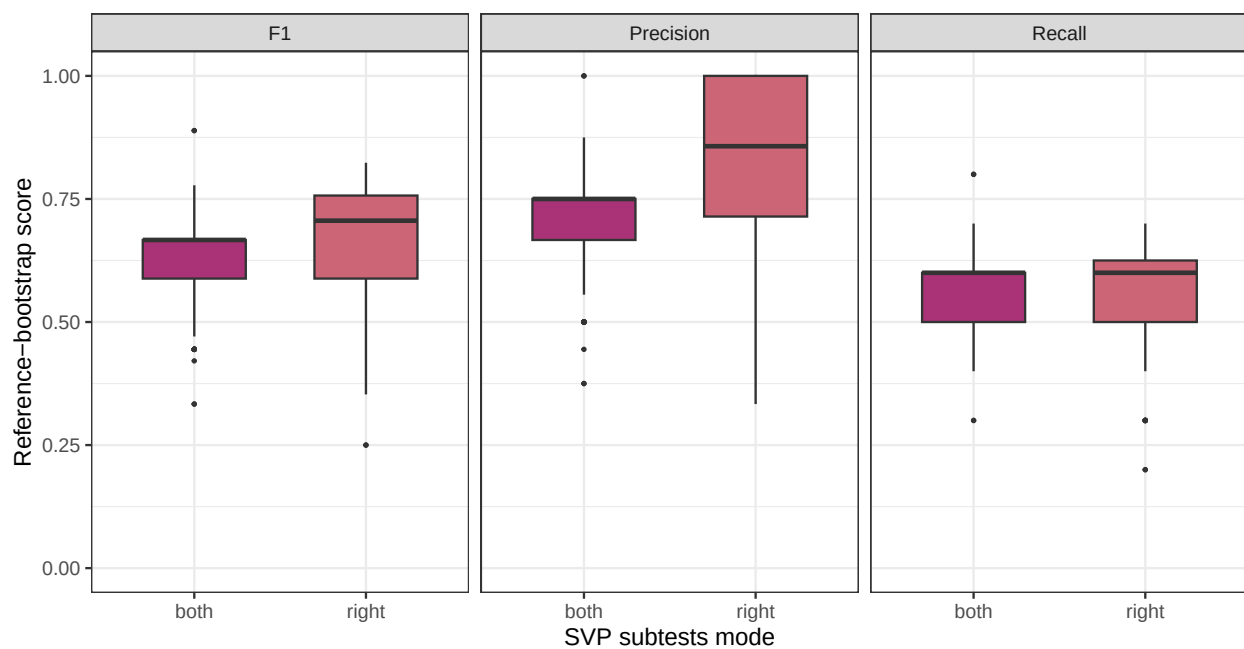
5 Mode and threshold sensitivity

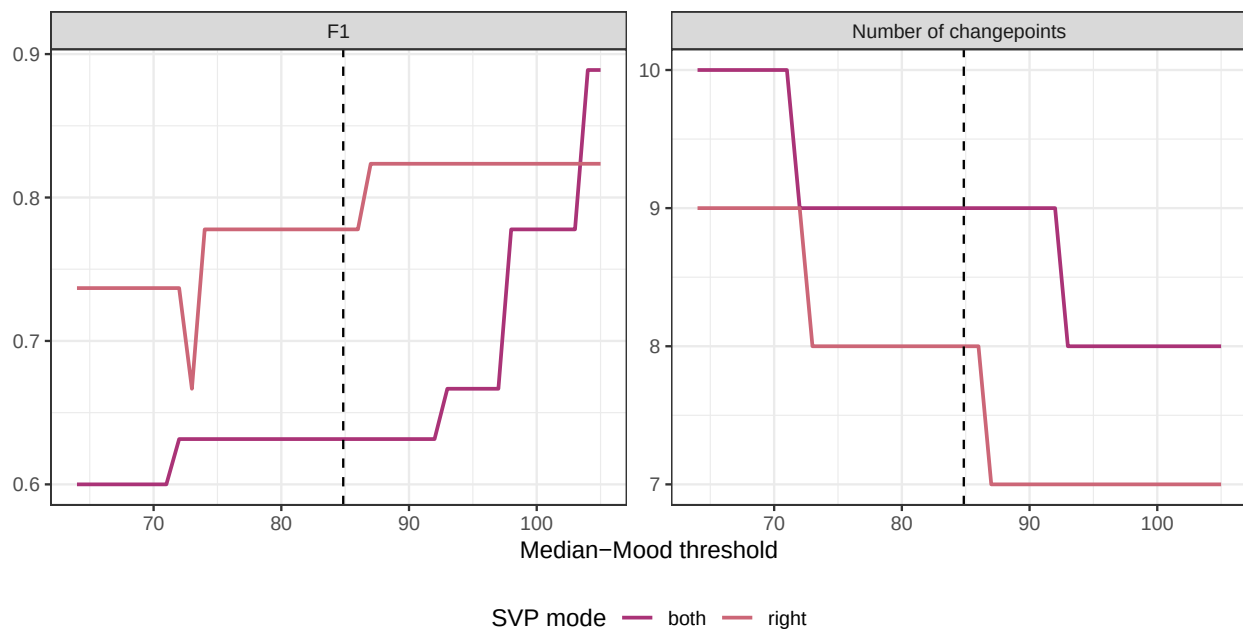
The signal bootstrap adds independently resampled residual blocks to the piecewise-median RFPOP signal. It is therefore a model-assisted comparison of the two SVP modes, not independent geological validation.

Table 5: Mode comparison over 500 signal-plus-residual bootstrap series.

mode	precision	recall	F1	mean CPs	reps
right	0.838	0.557	0.669	6.612	500
both	0.728	0.581	0.646	7.988	500

Across these replicates, `right` has mean F1 0.669, compared with 0.646 for `both` at the same threshold.





6 Interpretation and limitations

The calibrated Median-Mood **right** configuration is the defensible replacement for the current application settings: it removes the unknown oracle segment count and controls the whole-series null false-segmentation probability while retaining the main RFPOP transitions. The **both** mode can look closer to RFPOP at a lower threshold, but the null experiment shows that this comes from uncontrolled interior-subsegment multiplicity rather than better calibration.

This remains an empirical recommendation. RFPOP is not geological ground truth, and the bootstrap null is conditional on its fitted segmentation. The **right** result is also direction-dependent; the reverse-series comparison is reported in the CSV output. Exact pruning equivalence for the Median-Mood validity rule has not been proved. Finally, the Gaussian segment cost is only SVP's tie-breaker among partitions with the same number of segments; a robust tie-breaking cost is a separate possible extension.

7 Reproducibility

The analysis used `svpChange2` 0.2.0 from `/private/tmp/svpChange2-current/svpChange2`, `robseg` 2019.7.2, data length 4050, and data SHA-256 fingerprint `b95f02f50e829f2e7c453a49873f7329b3b76b88b4369fedff260bf209973fdf`. Calibration, validation, and signal-bootstrap seeds are 981000, 982000, and 983000, respectively.